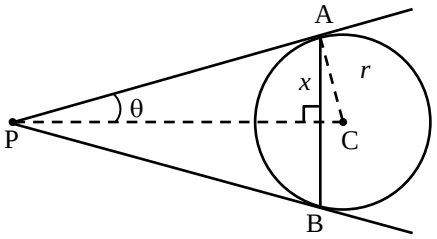
# MATHEMATICS

## SECTION – I

## Single Correct Choice Type

**1.** **(a)** $\tan\theta = \dfrac{r}{PA}$

Given $\dfrac{1}{r^2} + \dfrac{1}{PA^2} = \dfrac{1}{16}$




$\Rightarrow \quad \dfrac{\cot^2\theta + 1}{(PA)^2} = \dfrac{1}{16}$

$\Rightarrow \quad PA\sin\theta = 4 = x \quad \Rightarrow \quad 2x = 8$

**2.** **(b)** Coefficient of $x^2 = \dfrac{1}{2}\left[\left(1 + 2 + 2^2 ..... + 2^n\right)^2\right] - \left[1^2 + 2^2 + ..... \left(2^n\right)^2\right]$

$$= \frac{1}{2}\left[\left(\frac{2^{n+1}-1}{1}\right)^2 - \left(\frac{\left(2^2\right)^{n+1}-1}{3}\right)\right]$$

$$= \frac{1}{2}\left[\left(2^{n+1}-1\right)\left[\left(2^{n+1}-1\right) - \frac{2^{n+1}+1}{3}\right]\right]$$

$$= \frac{\left(2^{n+1}-1\right)\left(2^{n+1}-2\right)}{3}$$

**3.** **(c)** $f(x) = \left(4\sin^2 x - 1\right)^n \left(x^2 - x + 1\right)$

$f'(x) = n\left(4\sin^2 x - 1\right)^{n-1}(8\sin x\cos x)\left(x^2 - x + 1\right) + \left(4\sin^2 x - 1\right)^n (2x - 1)$

$= \left(4\sin^2 x - 1\right)^{n-1}[n\left(x^2 - x + 1\right).4\sin 2x + (2x - 1)\left(4\sin^2 x - 1\right)]$

If $f$ has local minimum at $x = \dfrac{\pi}{6}$, then

$f'\left(\dfrac{\pi}{6}\right) = 0$, $f'\left(\dfrac{\pi}{6}^-\right) < 0$ and $f'\left(\dfrac{\pi}{6}^+\right) > 0$,

which is possible if $(n-1)$ is odd i.e. $n$ even number.

**4.** **(c)** $I_2 = \dfrac{1}{3}\displaystyle\int_0^1 \frac{3x^2 dx}{e^{x^3}(2 - x^3)}$ put $t = x^3$

$I_2 = \dfrac{1}{3}\displaystyle\int_0^1 \frac{dt}{e^t(2-t)}$; $I_2 = \dfrac{1}{3}\displaystyle\int_0^1 \frac{dt}{e^{1-t}(1+t)}$

$\Rightarrow \quad \dfrac{I_1}{I_2} = 3e$

**5.** **(a)** $f'(x) = 3x^2 + 2ax + b$

$f(x)$ is increasing

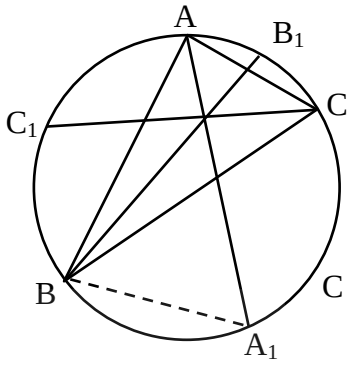
$\therefore$ $f'(x) \geq 0 \ \forall x$ where equality holds for discrete points.

$\therefore$ $D = (2a)^2 - 4 \times 3 \times b \leq 0 \ \Rightarrow \ a^2 - 3b \leq 0$

This is true for exactly 16 ordered pairs

$\therefore$ $\frac{16}{36} = \frac{4}{9}$.

**6.** **(a)** Join $BA_1$ as shown in diagram then




By sine rule for $\Delta ABA_1$

$AA_1 = 2\sin\left(B + \frac{A}{2}\right) = 2\cos\left(\frac{B}{2} - \frac{C}{2}\right)$

Thus $AA_1 \cos\frac{A}{2} = 2\cos\left(\frac{B-C}{2}\right)\cos\frac{A}{2} = \sin C + \sin B$

Similarly $BB_1 \cos\frac{B}{2} = \sin A + \sin C$

and $CC_1 \cos\frac{C}{2} = \sin A + \sin B$

So, $\frac{AA_1 \cos\frac{A}{2} + BB_1 \cos\frac{B}{2} + CC_1 \cos\frac{C}{2}}{\sin A + \sin B + \sin C} = 2$

## SECTION – II
### Multiple Correct Choice Type

**7.** **(b), (c), (d)**

Let $f(x) + \frac{a}{30} = 0$

where $f(x) = \frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} = \frac{(x-1)(x-2)(x-3)}{(x-1)(x-2)(x+4)} = \frac{x-3}{x+4}$

$x \neq 1, 2, -4$

Range of $f(x) = R - \left\{1, \frac{-2}{5}, \frac{-1}{6}\right\}$

$\therefore$ $\frac{a}{30} \neq -1, \frac{2}{5}, \frac{1}{6}$

$\Rightarrow$ $a \neq -30, 12, 5$

**8. (a), (b)**

Here for the parabola $y^2 = 8x$, $a$ = 2, Let $P(2t^2, 4t)$ be a point on parabola and $S(2, 0)$.

$SP = 4$

$\Rightarrow \quad a(1+t^2) = 4 \quad \Rightarrow \quad t = \pm 1$

$P$ = (2, 4) or (2, – 4)

$f(x + y) = f(x)\, f(y)$ ; $f(1) = 2$ ; $f(2) = 2^2$

$f(n) = 2^n$ ; $a_1 = 2$ ; $a_2 = 4$

$\therefore \quad P \equiv (a_1, a_2)$ or $(a_1, -a_2)$.

**9. (b), (d)**

$$\int \frac{dx}{1-\sin^4 x} = \frac{1}{2}\left(\int \frac{dx}{1+\sin^2 x} + \int \frac{dx}{1-\sin^2 x}\right)$$

$$= \frac{1}{2}\int \frac{dx}{\cos^2 + 2\sin^2 x} + \frac{1}{2}\int \frac{dx}{\cos^2 x} = \frac{1}{2}\int \frac{\sec^2 x}{1+2\tan^2 x}dx + \frac{1}{2}\int \sec^2 x dx$$

$$= \frac{1}{2\sqrt{2}}\tan^{-1}(\sqrt{2}\tan x) + \frac{1}{2}\tan x + C\ .$$

**10. (a), (b)**

Since $\vec{a}$ and $\vec{b}$ are unit vectors

$\therefore \quad |\vec{a} - \vec{b}| = \sqrt{(\vec{a} - \vec{b})^2} = 2|\sin\theta| < 1$

$|\sin\theta| < \frac{1}{2}$ ; $\theta \in \left(0, \frac{\pi}{6}\right)$ or $\left(\frac{5\pi}{6}, \pi\right]$.

**11. (a), (b), (c)**

$\alpha^{11} = 1 \Rightarrow \alpha^{11} - 1 = 0 \Rightarrow (\alpha - 1)(\alpha^{10} + \alpha^9 + ..... + \alpha + 1) = 0$

$\lambda = \alpha^6$ is a root of

$1 + x + x^2 + .....x^{10} = 0$ ..... (i)

$1 + \lambda + \lambda^2 + ..... + \lambda^{10} = 0$

$\Rightarrow \quad 1 + \lambda + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5 + \lambda + \lambda^2 + \lambda^3 + \lambda^4 + \lambda^5 = 0$

$\text{Re}(\lambda + \lambda^2 + ....\lambda^5) = -\frac{1}{2}$

$\beta, \beta^2, \beta^3 .....\beta^{10}$ are roots of (i)

$\therefore \quad (x - \beta)(x - \beta^2)(x - \beta^3).....(x - \beta^{10}) = 1 + x + x^2 + ..... + x^{10}$

Now put $x = \mu$

$(\mu - \beta)(\mu - \beta^2).....(\mu - \beta^{10}) = 0$ Put $x = i$

$(i - \beta)(i - \beta^2).....(i - \beta^{10}) = i$ .

## SECTION–III
## Reasoning Type

**12.** **(d)** $f(x)=[x]^2-\sqrt{\{x\}^2}$ is continuous at $x = 1$.

**13.** **(b)** If each of $x, y, z$ is less than $1/2$, then statement-1 is obviously true.

$\Rightarrow$ If one of $x, y, z$ is more than or equal to $\frac{1}{2}$, then their product is less than equal to zero hence, statement-1 still remain true.

**14.** **(b)**

**15.** **(d)** $A = \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$
Number of different rational numbers = $10p_2 + 1$,

(1 corresponding to numbers like $\frac{2}{2}$ or $\frac{3}{3}$)

$=10 \times 9 + 1 = 91$

**16.** **(d)** $\min\left(\sqrt{3},\ a^2-4a+5\right)\in[1,\ \sqrt{3}]$ and $(\sin x-2\cos x)\in[-\sqrt{5},\ \sqrt{5}]$
So there exist infinite solutions.

## SECTION–IV
## Matrix–Match Type

**1.** $f(x)>1 \Rightarrow x>0$
$f(x)<-1 \Rightarrow x<0$
$f(x)>3 \Rightarrow 0<x<\frac{1}{2}$
$f(x)<-3 \Rightarrow -\frac{1}{2}<x<0$

**A-1; B-1, 3; C-2, 4; D-2**

**2.** **A-1, 4, 5** Total number of cases $= 9!$
and number of favourable case are $= {}^9C_3 \times 6!$
Required probability $= \frac{{}^9C_3 \times 6!}{9!} = \frac{1}{6}$

**B-2, 3** Required probability $= \frac{{}^9C_5 \times {}^4C_2 \times 4!}{9!} = \frac{1}{20}$

**C-2, 3** Required probability $= \frac{{}^8C_5 \times {}^4C_2 \times 3!}{8!} = \frac{1}{20}$

**D-1, 4, 5** Required probability $= \dfrac{{}^{9}C_3 \times {}^{6}C_3}{{}^{9}C_3 \times {}^{6}C_3 \times 3!} = \dfrac{1}{6}$

## SECTION–V
## Integer Answers Type

**1.** Solving the equation of the lines we get $z = -\bar{z} \Rightarrow z = i$

$|\alpha - i| = 2$ ; $\alpha = 2e^{i\theta} + i$ , put it in the equation of the second line, we get

$\cos\theta - \sin\theta = 0$

$\Rightarrow \quad \theta = \dfrac{\pi}{4}$

$\alpha = i \pm 2e^{\frac{i\pi}{4}}$

$\therefore \quad x = \pm\sqrt{2}$

$\Rightarrow \quad [|x|] = 1$

**Ans. 1**

**2.** Let $P$ be $(5\cos\theta, 4\sin\theta)$; $Q$ be $(-5\sin\theta, 4\cos\theta)$

Equation of tangent at $P$ $\dfrac{x}{5}\cos\theta + \dfrac{y}{4}\sin\theta = 1$ ..... (i)

Equation of tangent at $Q$ $-\dfrac{x}{5}\sin\theta + \dfrac{y}{4}\cos\theta = 1$ ..... (ii)

Solving (i) and (ii) $\Rightarrow$ $R \equiv (5(\cos\theta - \sin\theta),\ 4(\sin\theta + \cos\theta))$

$\therefore \quad m : n$ is 1 : 1

$\Rightarrow \quad m + n = 2$

**Alternate:**

Let $P(5, 0)$; $Q(0, 4)$

$\Rightarrow \quad R(5, 4)$

Intersection of $CR$ and $PQ$ is $\left(\dfrac{5}{2}, 2\right)$, which is mid-point of $CR$

$\Rightarrow \ m:n = 1:1 \ \Rightarrow \ m + n = 2$

**Ans. 2**

**3.** $I = \int_0^1 \underset{\text{I}}{\left(1 - x^4\right)^7} \cdot \underset{\text{II}}{1}\, dx$

$= [x(1 - x^4)^7]_0^1 + 7 \times 4\int_0^1 x(1 - x^4)^6 x^3 dx$

$= -28\int_0^1 (1 - x^4)(1 - x^4)^6\, dx + 28\int_0^1 (1 - x^4)^6 dx = -28I + 28\int_0^1 (1 - x^4)^6 dx$

$29I = 28\int_0^1 (1 - x^4) dx$

$$\frac{29\int_0^1 (1-x^4)^7\,dx}{4\int_0^1 (1-x^4)^6\,dx} = 7$$

**Ans. 7**

**4.** $f'(x) = -a\pi\sin(\pi x)$

$\Rightarrow \quad f'\left(\frac{1}{2}\right) = -a\pi\sin\frac{\pi}{2} = -a\pi = \pi \;\Rightarrow\; a = -1$

$$\int (a\cos\pi x + b)dx = \left(\frac{a\sin\pi x}{\pi} + bx\right)_{1/2}^{3/2} = \left(\frac{-a}{\pi} + \frac{3b}{2}\right) - \left(\frac{a}{\pi} + \frac{b}{2}\right)$$

$$\Rightarrow \quad -\frac{2a}{\pi} + b = \frac{2}{\pi} + 1 \;\Rightarrow\; b = 1$$

So, $\frac{-12}{\pi}\left(\sin^{-1}(-1) + \cos^{-1}1\right) = \frac{-12}{\pi}\left(-\frac{\pi}{2} + 0\right) = 6$

**Ans. 6**

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